

Object Detection Algorithm Based on Lightweight YOLOv4 for UAV

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Abstract—With the development of artificial intelligence technology, the mission scope of UAV has gradually expanded from the traditional image acquisition of high-altitude target area to aerial photography of cities, mapping regions, forest inspection, plant protection monitoring, rescue missions, fire rescue and other aspects. In recent years, the industrial development of UAV has reduced the cost of UAV, improved the performance of camera, stability and other aspects, and more UAV has been put into real life. Among them, the research on object detection in UAV images has become a hot topic, and a large number of excellent algorithms have emerged, aiming at improving the accuracy and speed of object detection. The object detection of UAV is faced with the following problems: small detected targets in the field of vision, weak computing power of the embedded system, weak real-time detection, etc. By comparing some classical object detection algorithms, using YOLOv4 as the basic network model, the structure of the network and the algorithm of image prediction box selection are improved. Then, a lightweight YOLOv4 object detection algorithm for small target is proposed. It can not only improve the speed and accuracy of object detection for the UAV, but also apply to mobile or embedded platform of the UAV. What's more, the realization of real-time object detection greatly improves the UAV's scene understanding ability and operational efficiency.

Keywords—object detection; UAV; YOLOv4; lightweight

I. INTRODUCTION

UAV is a new technology product in the information age, which involves data collection, intelligent control, information processing, communication and other related technologies. It has the advantages of flexibility, maneuverability and multiple application scenarios. Especially since the outbreak of COVID-19, the UAV have gradually become an important guarantee in social life. The UAV is different from manned aircraft. It needs to perform relevant tasks in a weak human-computer interactive environment. This undoubtedly requires intelligent interpretation and processing ability of the scene. Moreover, the improvement of understanding of the scene must be based on object detection. And, not only to ensure a good accuracy, but also to carry out real-time detection of the scene, in order to

ensure better application in practical engineering. The application of deep learning technology in the object detection algorithm, and selecting a deeper convolutional neural network, which improves the model complexity and helps improve the accuracy [1]. However, most complex models have these problems, such as large number of parameters, large model, etc. In addition, these models have high requirements on computing performance and storage devices. These models cannot be deployed on small devices such as the UAVs, which have far less computing power than personal computer. Therefore, it is necessary to compress and accelerate the algorithm. The compression model leads to the destruction of the features extracted from the convolution layer, resulting in the decrease of the accuracy. The lightweight detection model adopts more efficient convolution methods, such as separable convolution and deeply separable convolution. It helps to reduce model parameters and ensure the performance of the model. At the same time, a multi-scale path aggregation module is needed because of its poor detection effect on small targets. To solve the above problems, an improved lightweight detection algorithm based on YOLOv4 was proposed, which is suitable for UAV.

II. SELECTION OF OBJECT DETECTION MODEL FOR UAV

Object detection is a basic and important problem in computer vision. The task is to segment and recognize objects according to their statistical and geometric characteristics. The purpose of object detection is to extract the specified target features from digital images through algorithms, including the length and width information, location information, category information and classification confidence. Now the classical object detection models include two-stage and one-stage. Two-stage object detection is to generate a series of region proposals as samples by the algorithm, and then conduct sample classification through the convolutional neural network, including fast-RCNN [2], faster-RCNN [3], R-FCN [4] and other models, which has high accuracy of object detection and positioning accuracy. The one-stage object detection model does not generate region proposals and directly transforms the

problem of target bounding box location into regression problem. It mainly includes SSD [5], YOLOv1, YOLOv2, YOLOv3, YOLOv4[6], etc., with a small amount of calculation and relatively fast algorithm running speed. PASCAL

VOC2007 and 2012 public data sets were used to conduct comparative tests on the above models, and the experimental results are shown in Table I.

TABLE I. PERFORMANCE COMPARISON OF ONE-STAGE AND TWO-STAGE ALGORITHMS

Model	Basic Structure	VOC2007 (m A P / %)	VOC2012 (m A P / %)	GPU time (ms)
<i>Fast-RCNN</i>	<i>VGG-16</i>	70.0	68.4	2532.72
<i>Faster-RCNN</i>	<i>VGG-16</i>	73.2	70.4	1692.51
<i>R-FCN</i>	<i>ResNet-101</i>	80.5	77.6	1422.92
<i>SSD</i>	<i>VGG-16</i>	77.2	75.8	27.62
<i>YOLOv1</i>	<i>GoogLeNet</i>	63.4	57.9	182.92
<i>YOLOv2</i>	<i>DarkNet-19</i>	76.8	73.4	59.10
<i>YOLOv3</i>	<i>DarkNet-53</i>	79.1	77.2	38.54
<i>YOLOv4</i>	<i>CSPDarknet-53</i>	85.9	86.2	27.20

As can be seen from the evaluation results in Table 1, in terms of detection accuracy, YOLOv4 has the highest accuracy and the best real-time performance. Therefore, this paper chooses YOLOv4 (You Only Look Once) model as the basic model.

YOLOv4 model regards object detection as an end-to-end regression problem, and the final result is obtained by directly calculating the classification result and object position coordinate of the input image. The model includes four parts: Input, Backbone for feature extraction, Neck for feature fusion and Head for classification and regression. Input mainly includes mosaic data enhancement, Cross Small Batch Standardization, and Self-Adversarial Training (SAT). Backbone not only uses CSPDarkNet53, but also uses Mish activation function, which reduces computation and ensures algorithm accuracy. In Neck, a Path Aggregation Network (PANet) module containing Space Pyramid Pool (SPP) module is adopted. The SPP module transforms the input feature graph into feature graph of different scales by means of maximum pooling, and then concat the feature graph of different scales and the original feature graph and outputs them, which is beneficial to expand the sensibility field of convolution. Head follows YOLOv3's multi-scale prediction method. In the training process, lable smoothing and CLOU loss function are used to improve the accuracy of the model. It can be said that YOLOv4 model that based on the combination of the above features achieves the balance between detection speed and accuracy.

Object detection needs to be adjusted and optimized for different situations in different application scenarios. It is most important to improve the real time detectability of object detection for UAV. Although YOLOv4 has reached a good balance in accuracy and speed, due to its backbone feature extraction network CSPDarkNet53 uses a large number of stacked residual structures. As a result, the amount of model parameters and calculations are too huge to apply to mobile or embedded platforms. At the same time, due to the small number of pixels occupied by the target in the field of vision of UAV, the number of targets is large and dense, so the model

needs to improve, in order to improve the effect on the small targets of object detection for the UAV. Therefore, a real-time object detection based on lightweight YOLOv4 is proposed in this paper.

III. OBJECT DETECTION ALGORITHM BASED ON IMPROVED YOLOV4 FOR UAV

A. The improvement of network structure

MobileNet convolutional neural network has many advantages, including extremely low number of parameters and computation. It is not only more suitable for deployment on embedded system devices with low computational load, but also can better balance the accuracy of detection model and the efficiency of operation. Based on the YOLOv4 object detection algorithm, MobileNetV3 is used to replace the original feature extraction network CSPDarkNet53 network. In order to further reduce the number of parameters and model size, achieve the purpose of improving the detection speed. The SPP+PAN+YOLO Head structure of YOLOv4 is still used in neck and head, and part of 3*3 ordinary convolution in the original PANet is replaced by deep separable convolution. After using depth-separable convolution, the amount of computation and parameters will be reduced to about 1/4 of the YOLOv4, which significantly reduces the model size and improves the detection speed. The overall network structure of improved YOLOv4 is shown in Fig. 1.

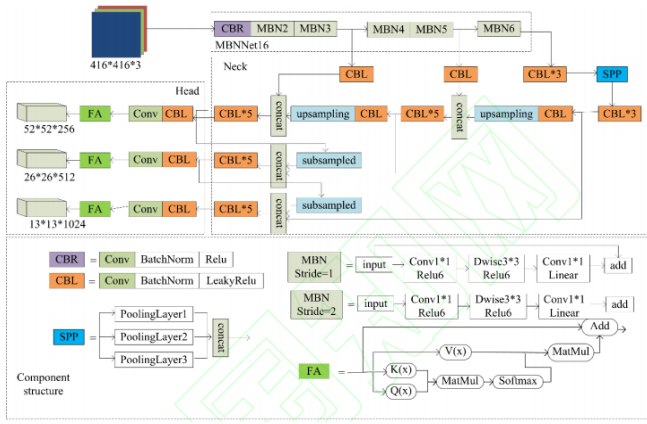


Figure 1. The network structure of lightweight YOLOv4

B. The improvement of image prediction boxes screening algorithm

When the number of generated anchor boxes is large, the same target may be detected repeatedly, which leads to many similar prediction boxes. YOLOv4 uses Dou-NMS. Although the method solves the error suppression of the dense case that exists in the Non-maximum Suppression, it does not solve the defect that the box with higher default confidence score of NMS is more accurate. Because classification is not directly related to regression tasks, bounding boxes with high confidence scores are not always more reliable than boxes with low confidence scores. To remedy this deficiency, DIOU-NMS of YOLOv4 is replaced by Softer-NMS [7]. Firstly, the confidence of overlapping detection frames is reduced rather than filtered directly. Then, a new positioning confidence prediction is added on this basis to make the frame position with high classification confidence more accurate. In conclusion, the performance of object detection can be improved effectively.

1) Non-maximum Suppression

The formula is as shown in (1):

$$S_i = S_i e^{-\frac{IOU(M, b_i)}{\theta}} \quad \forall b_i \in D \quad (1)$$

Where S_i is the score of each prediction box; M is the boundary box with the highest score; b_i is a series of prediction boxes for network prediction; D is the set of a series of prediction boxes retained after processing by the Non-maximum Suppression, and D is initially set as an empty set. The value range of θ is (0,1), and 0.2 is taken in this paper.

According to Formula (1), the larger the IOU of b_i and M is, the more severe the decrease of the score S_i of b_i will be, thus reducing its confidence and not directly setting it to 0.

2) Position confidence

Assuming that the four coordinate values of the boundary box are independent from each other, univariate Gaussian distribution is adopted to predict position confidence, and the formula is shown in (2):

$$P_{\theta}(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-x_g)^2}{2\sigma^2}} \quad (2)$$

Where, θ is the set of learnable parameters; x_g is the estimated boundary box position. Standard deviation σ is used to measure the uncertainty of prediction. When $\sigma \rightarrow 0$, it indicates that the network has a high level of confidence in the predicted position. GT boundary box confidence can also be expressed by gaussian distribution. When $\sigma \rightarrow 0$, it becomes the position of x_g GT boundary box of the Dirac function $P_D(x) = \sigma(x - x_g)$.

KL loss function: The objective of target positioning is to estimate parameter θ and minimize the KL divergence between $P_{\theta}(x)$ and $P_D(x)$ of N samples. The publicity is as shown in (3):

$$\hat{\theta} = \arg\theta \min \frac{1}{N} \sum D_{KL}[P_D(x) \parallel P_{\theta}(x)] \quad (3)$$

KL divergence is adopted as regression loss function, and for a single sample, the formula is shown as (4):

$$L_{reg} = D_{KL}[P_D(x) \parallel P_{\theta}(x)] = \int P_D(x) \log \frac{P_D(x)}{P_{\theta}(x)} dx \quad (4)$$

According to equation (4), when x_g prediction is inaccurate, greater σ^2 of network prediction results in less L_{reg} . $\theta \log(2\pi) / 2$ and $H[P_D(x)]$ has nothing to do with estimated parameters θ , therefore $L_{reg} \propto \frac{(x_g - x_g)^2}{2\sigma^2} + \frac{1}{2} \log(\sigma^2)$.

In order to avoid gradient explosion, $\alpha = \log(\sigma^2)$ rather than α is predicted in the actual network. Therefore, the final deduced loss function is shown in (5):

$$L_{reg} = e^{-\alpha} \left(|x_g - x_g| - \frac{1}{2} \right) + \frac{1}{2} \alpha \quad (5)$$

On the basis of NMS, the Softer-NMS does not directly remove the detection frames with high overlap, but reduces the classification confidence score of the overlapping detection frames. This method can solve the problem that the boundary boxes with low scores are suppressed when the objects are dense. In addition, with the increase of position confidence prediction, the border position with high classification confidence becomes more accurate. The Softer-NMS has a good effect on achieving significant performance improvement and improving the positioning accuracy of object detection.

In addition, the self-attention can also be added in the head part. The self-attention can be used to estimate the relevant information from the input feature image and build the global dependencies among all feature pixels in the feature image. Then achieve more effective object detection for different regions.

IV. SIMULATION EXPERIMENT AND RESULT ANALYSIS

A. Experimental environment

Enumerating the experimental environment of this paper from hardware and software, other dependent function libraries are omitted. The workstation is configured with Core I9-10900K CPU and Nvidia Geforce GTX 3080 GPU. The deep learning framework is PyTorch-GPU=1.7. CUDA10.1 and Cudnn7.6.0 deep learning accelerators are installed. In addition, to verify the applicability of the model in UAV flight, the model deployment and experimental results were analyzed on embedded systems Nvidia Jetson TX2 and Raspberry Pi 4B.

B. Experimental data

In this experiment, PASCAL VOC containing 20 categories were selected as the model training and testing data set. The ratio of validation subset and training subset was 1:9. In order to reduce the influence of extra factors on recognition, data enhancement is performed on the original data set. The weight decay factor of the optimizer was removed, namely weight decay=0.

C. Evaluation index

To evaluate the performance of object detection algorithm, average accuracy (**AP**) is usually used to measure the detection effect of single target, and average accuracy (**mAP**) is used to measure the detection effect of multiple targets. In addition, the evaluation of model size is usually measured by parameter and computational amount. The smaller the number of parameters and calculation, the lighter the model. What's more, FPS is usually used to evaluate detection speed. FPS refers to the number of frames detected per second. The larger the FPS is, the faster the algorithm detects. In this paper, **AP**, **mAP**, FPS, parameter number and model size were selected as evaluation indexes of the model. **AP** is the area of P-R curve, higher **AP** means better detection performance, and **mAP** is the average value of **AP** about all categories. Relevant calculation formulas are as follows: (6-10)

$$P = \frac{TP}{TP+FP} \quad (6)$$

$$R = \frac{TP}{TP+FN} \quad (7)$$

$$F_1 = \frac{2PR}{P+R} \quad (8)$$

$$AP = \int_0^1 P(R) dR \quad (9)$$

$$mAP = \sum_{i=1}^N AP_i / N \quad (10)$$

Where, **TP** is the number of real samples, indicating the number of samples that the detected target category is consistent with the real category; **FP** is a false positive sample, indicating the number of samples that the detected target

category is inconsistent with the real category; **FN** is false negative sample, indicating the number of samples that the real target exists but is not detected by the network model; **AP** is the average accuracy; **N** is the number of targets for all categories.

D. Experimental results and analysis

After the self-attention is added to YOLOv4 algorithm, the accuracy is improved to some extent, and the speed is slightly reduced. Lightweight network MobileNetV3 was used for feature extraction, and improved Non-maximum Suppression is also used for post-processing. This results in a small decrease in the accuracy 83.62%. However, the speed has been greatly improved, and the FPS reaches 61, which is 1.8 times faster than the speed of the original YOLOv4. On the basis of YOLOv4, self-attention and Softer-NMS are introduced. So that, the accuracy reaches 86.73% and the FPS reaches 34. Compared with the original YOLOv4, the accuracy and the speed all increase. The algorithm in this paper uses the above three improved methods at the same time, which makes the accuracy reach 85.65%. Compared with the original YOLOv4, the FPS reaches 53, which is 20 higher. It can be clearly seen that each improved method plays an important role in the object detection algorithm, which is complementary and indispensable. The values of mAP and FPS in the model comparison at each stage of the algorithm optimization process in this paper are shown in Table II.

TABLE II. COMPARISON OF MODELS IN EACH STAGE OF THE ALGORITHM OPTIMIZATION PROCESS IN THIS PAPER

Model	mAP/%	FPS
<i>YOLOv4</i>	86.32	33
<i>YOLOv4+MobileNetV3</i>	83.08	57
<i>YOLOv4+MobileNetV3+ Softer-NMS</i>	83.62	61
<i>YOLOv4+ +SE+Softer-NMS</i>	86.73	34
<i>YOLOv4+SE+Softer-NMS+MobileNetV3</i>	85.65	53

The results of contrast between YOLOv4 and the model proposed in this paper applied in UAV embedded systems Nvidia Jetson TX2 and Raspberry Pi 4B are shown in Table III.

TABLE III. FPS COMPARISON OF DIFFERENT METHODS OF UAVS

Model	FPS	
	<i>Nvidia Jetson TX2</i>	<i>Raspberry Pi 4B</i>
<i>YOLOv4</i>	5.6	2.8
<i>This paper model</i>	23.8	9.6

The results show that the improved model can improve the performance of object detection.

V. CONCLUSION

With the deep learning development, in the past few years, object detection algorithm has been highly concerned at home and abroad, and developed rapidly. Although relevant algorithm optimization has achieved good results, there is still room for optimization and improvement. From the perspective

of the main research directions in recent years, these researches mainly focus on seeking the optimal convergence point between object detection accuracy and object detection speed. Although the object detection based on YOLOv4 algorithm has not been proposed for a long time, the research trend is more in line with the development of UAV. It also points out the direction for the improvement and optimization of object detection algorithm for UAV.

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